

The Vocabulary of Gratian's *Decretum*: Change Over Time

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The textual history of Gratian's Decretum is now understood well enough to enable researchers to compare different recensions of the text to identify evolutionary changes in its doctrine. Specific developments can be identified by the close reading of well-chosen selections from the text. However, a systematic search for such doctrinal developments is best conducted with computational assistance, taking advantage of recent advances in the application of large language models to natural language processing tasks. Using the PIE lemmatizer in conjunction with the LASLA Latin models, it is possible to lemmatize samples reflecting different compositional stages of sections of the Decretum and to compare the results by isolating lemmas unique to specific stages in the development of the text. Distinctive vocabularies of different compositional stages revealed by this method then provide a roadmap for close reading.

This paper presents work that is part of a larger project to enhance the effectiveness of close reading of medieval texts, Gratian's *Decretum* in particular, using computational assistance. The Digital Humanities community uses the term “distant reading” to describe this kind of computational assistance, emphasizing both its connection to, and its contrast with, traditional close reading. Each project that uses a distant or machine reading approach aims to answer different questions and therefore uses different tools and techniques.

The original project from which the current work grew was concerned with the question of the authorship of the case statements and the first- and second-recension *dicta* in Gratian's *Decretum*, including the *dicta* in *de penitentia*. It employed stylometric authorship analysis using

a statistical technique from computational linguistics—principal component analysis of the frequencies of commonly occurring function words—to obtain its results. Its conclusion was that the case statements were written by a single author who was not the author of the *dicta* either in the first or second recensions or in *de penitentia*. Results from stylometric analysis for authorship of the *dicta*, however, were not consistent with either the one-author theory championed by Kenneth Pennington or the two-author theory championed by Anders Winroth.¹² Instead, the results suggested, but did not conclusively prove, that both the first- and the second-recension *dicta* were the work of multiple authors.³

In contrast, the current work concerns changes in the teaching or doctrine of the *Decretum* between the first and second recensions as indicated by the use of distinctive vocabulary in the *dicta* of the two recensions. It employs another computational linguistic technique, lemmatization, using the PIE lemmatizer and a large language model (LLM) trained on the corpus of lemmatized and tagged Latin text created by the Laboratoire d'Analyse Statistique des

¹ Kenneth Pennington, “The Biography of Gratian, the Father of Canon Law,” in *A Service Beyond All Recompense*, ed. Kurt Martens (Catholic University of America Press, 2018), 359–92.

² Anders Winroth, *The Making of Gratian's Decretum* (Cambridge: Cambridge University Press, 2000), 175–92.

³ Paul Evans, “Distant Reading of Gratian's Decretum” (PhD thesis, The Catholic University of America, 2022), <https://jstor.org/stable/community.38760479>.

Langues Anciennes (LASLA) at the University of Liège.⁴⁵ The two projects share a common data set. The original project relied on a very carefully prepared data set of the texts of the case statements and the first- and second-recension *dicta*.⁶ The current work uses the same data set. The overarching goal of the two projects is the same—to use computational distant reading techniques to enhance close readings of the *Decretum*—but the two projects attempt to answer different questions using different tools and techniques.

This is not my first attempt to make progress on the problem of computationally identifying topics added to the *dicta* between the first and second recensions. Around 2012, there was tremendous enthusiasm in the Digital Humanities community for a technique called unsupervised topic modeling and in particular for a topic-modeling tool called the MACHINE

⁴ Enrique Manjavacas, Ákos Kádár, and Mike Kestemont, “Improving Lemmatization of Non-Standard Languages with Joint Learning,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (Minneapolis, Minnesota: Association for Computational Linguistics, 2019), 1493–503.

⁵ Philippe Verkerk et al., “L.A.S.L.A. And Collatinus: A Convergence in Lexica,” *Studi e Saggi Linguistici* 58 (September 2020): 95–120.

⁶ This research could usefully be expanded to include the rubrics and canons, and I have made preparations to do so. The work required to expand the corpus to include the canons themselves would require grant funding.

Learning for Language Toolkit (MALLET).⁷ Inspired by Pennington's observation that most passages in the *Decretum* dealing with the legal status of Jews, particularly those dealing with forced conversion, were introduced only in the second recension,⁸ I hoped to use MALLET to identify other new topics added in the second recension. The approach was to topic model the first- and second-recension *dicta* together, then separately, in order to show which topics were left when the first recension topics were subtracted. (It would not be enough to topic model just the second-recension *dicta* because many of the topics present in the second recension *dicta* are also present in the first-recension *dicta*.) This approach was simple in concept but prohibitively difficult in practice, for two reasons: first, because of the difficulty in determining the number of topics to look for, a necessary precondition for unsupervised topic modeling, and, second, because there was no obvious way to subtract topics.

For the discussion that follows, I am defining "first-recension *dicta*" as the text of the *dicta* listed in the appendix of Winroth's *The Making of Gratian's Decretum*, and I am defining "second-recension *dicta*" as the words in the *dicta* as they appear in the Friedberg edition when the words in the *dicta* in Winroth's appendix have been taken away. D.54 d.p.c.23 is a good example, Winroth's appendix indicates that only the first sentence of the *dictum* appears in the first

⁷ Andrew Kachites McCallum, "MALLET: A Machine Learning for Language Toolkit" (2002).

⁸ Kenneth Pennington, "The Law's Violence Against Medieval and Early Modern Jews," *Rivista Internazionale Di Diritto Comune* 23 (2013): 23–44; and Kenneth Pennington, "Gratian and the Jews," *Bulletin of Medieval Canon Law* 31 (2014): 111–24.

recension. Therefore, the first sentence of the *dictum* is assigned to the first recension text sample:

Ecce, quomodo serui ad clericatum ualeant assumi, uel quomodo non admittantur.

Liberti quoque non sunt promouendi ad clerum, nisi ab obsequiis sui patroni fuerint absoluti. Unde in Concilio Eliberitano:

The remainder of the *dictum* is assigned to the second recension text sample:

Ecce, quomodo serui ad clericatum ualeant assumi, uel quomodo non admittantur.

Liberti quoque non sunt promouendi ad clerum, nisi ab obsequiis sui patroni fuerint absoluti. Unde in Concilio Eliberitano:

Using this definition, the first- and second-recension *dicta* contain 56,713 and 14,255 words respectively.

Once it became clear that unsupervised topic modeling using MALLET would not be an effective way to identify topics added to Gratian's *Decretum* between the first and second recensions, the most promising alternative approach to the problem appeared to be using lemmatization to identify distinctive vocabulary as an indicator pointing to new ideas added between the first and second recensions. When working in a highly inflected language like Latin, using words as the indicators pointing to corresponding ideas is not sufficiently precise. As an example to be examined more closely later in this paper, the noun *calumnia* has seven unique declined forms. A regular first conjugation deponent verb like *calumnior*, *calumniari*, *calumniatus* has 120 conjugated forms, approximately eighty of which are unique, not including participial forms. Therefore, if we want to use distinctive vocabulary as a basis for determining

whether or not an idea or topic is present in a Latin text, we need to lemmatize every word form we encounter—that is, reduce it to its dictionary headword or *lemma*. Once text samples for the first- and second recension *dicta* have been reduced to corresponding lists of lemmas, those lists can be compared to generate three further lists, (i) of lemmas that appear in both the first- and second-recension *dicta*, (ii) of lemmas that are unique to the first-recension *dicta*, and (iii) of lemmas that are unique to the second-recension *dicta*. It is the list of lemmas unique to the second-recension *dicta* that is relevant to the problem of topics added to the *Decretum* in the second recension.

The results of initial experiments with the Classical Language Toolkit (CLTK), built on top of the Python Natural Language Toolkit (NLTK) and the best lemmatization tool available at the time, were not encouraging.⁹ The first- and second-recension *dicta* might reasonably be expected to include a few hundred unique lemmas, but CLTK reported many thousands (over four thousand for the first-recension *dicta* alone), the overwhelming majority of which were false positives. Clearly, lemmatization was not ready for the purpose of this project, and that remained the case for many years, from 2014 through 2020.

In 2021, Mike Kestemont made me aware of the PIE lemmatizer. Kestemont is a researcher at the University of Antwerp specializing in medieval Latin and Middle Dutch literature and a

⁹ Python is a widely-used general-purpose programming language. According to one frequently-cited industry metric, the [TIOBE Index](#), Python is the most popular programming language worldwide as of June 2026. Python provides powerful features for performing operations on textual data.

leading figure in the field of computational text analysis. PIE is not an application or program—the user does not simply type a command or click a button and get lemmatized text as output. Instead, PIE and PIE Extended¹⁰ are a collection of libraries, packages, and toolkits that provide an extremely versatile set of software building blocks that can be called upon to perform a wide range of natural language processing functions, like part-of-speech tagging or lemmatization, in a Python program.¹¹ They are based on large language models (LLMs) trained using machine learning techniques on annotated corpora of texts in the target language. In this case, PIE uses a model trained on the LASLA corpus of 2.5 million words or tokens of classical Latin, each annotated with lemma, part of speech, and other morphological and syntactic information. Once the PIE lemmatization environment had been set up, I wrote a Python program that used PIE first to create separate lists of every lemma found in the first- and second-recension *dicta* and then to

¹⁰ Thibault Clérice, “Pie Extended, an Extension for Pie with Pre-Processing and Post-Processing” (Zenodo, June 2020), <https://doi.org/10.5281/zenodo.3883589>.

¹¹ I would like to acknowledge Jake Bayon, an undergraduate Computer Science student at the University of San Diego, who set up the PIE lemmatization environment as an independent study project with me during the Spring 2024 semester and who learned something about Gratian in the process. PIE can only be installed with the 2019 Python 3.8 release—the current release as of June 2026 is Python 3.14.

compare the two lists to identify lemmas that appear only in the second-recension *dicta*. The program produced a list of 725 unique lemmas present only in second-recension *dicta*.¹²

An understanding of whether an idea or topic is present in, or absent from, a selection of text can almost never be arrived at based on the presence or absence of a single lemma. Instead, human, as opposed to machine, readers must look for the presence of families of related lemmas to indicate the presence of an idea or topic in a selection of text. In reviewing the list of the 725 lemmas unique to the second-recension *dicta*, one such family, all related to the concept of calumny, stood out. This family will be the exemplar of what new computational techniques are able to reveal about the evolution of the text and ideas of the *Decretum*.

Calumny is a promising lead because we know that it was a topic to which canonists gave considerable attention before and during the classical period in the history of medieval canon law between 1140 and 1234. They concentrated their interest on the oath of non-calumniation (*de calumnia vitanda*), which came to be required of almost all litigants at the onset of legal proceedings in ecclesiastical courts. The oath had been a normal part of Roman civil procedure in the age of Justinian but disappeared from use in the West along with the rest of Justinianic Roman law in the Early Middle Ages. It reemerged in the West in the second quarter of the eleventh century. For example, the judge in a 1029 case from the bishop's court in Ravenna asked first the plaintiff's notary on behalf of his client and then the defendant to take the oath. Both declined to do so. It is noteworthy that this episode took place almost half a century before

¹² The 728 lines of program output included three numbers, which I discarded. The complete list is available at <https://github.com/decretist/ICMCL17/blob/main/results/lemmas.txt>.

the 1076 Placitum of Marturi (Poggibonsi), usually taken as signaling the revival of Justinianic Roman law in the West. By the second quarter of the twelfth century, use of the oath was widespread enough to make it an active topic of discussion during the reigns of Honorius II (1124-30), Innocent II (1130-1143), and Eugenius III (1145-1153). The contested issues were (i) who should be required to take the oath (some litigants, all litigants, advocates on behalf of litigants, or advocates as well as litigants), and (ii) the kinds of cases in which litigants or advocates or both should be required to take the oath (Eugenius treated cases concerning tithes, possession of churches, and “spiritual affairs” as exceptions to a general requirement to take the oath). The general trend over the course of the second half of the century was in the direction of extending the requirement and “[b]y the close of the twelfth century the calumny oath had become a normal element of canonical civil procedure and might be required in most contested cases, at least if one of the parties insisted upon it.”¹³

The idea of calumny is used in two distinct senses in the *Decretum*. The first is general and biblical, and is derived from sayings attributed to Jesus in which a relationship between calumny and the Mosaic prohibition against bearing false witness is implied though not directly stated. This general and biblical sense is also the one used in many of Gratian’s patristic material sources. The second is specific and legal, and is derived from Justinianic Roman law. Where the idea of calumny is invoked in the *dicta* of the first recension of the *Decretum*, it is exclusively in

¹³ James A. Brundage, “The Calumny Oath and Ethical Ideals of Canonical Advocates,” in *The Profession and Practice of Medieval Canon Law*, Variorum Collected Studies Series CS797 (Aldershot: Ashgate Variorum, 2004), 795–99.

the first sense; where it is invoked in the *dicta* of the second recension, it is predominantly, though not exclusively, in the second sense. Interestingly, Gratian does not discuss the oath *de calumnia vitanda* in either the first or the second recensions.

We should expect to see at least three Latin lemmas associated with the concept of calumny: the deponent verb *calumnior*, *calumniari*, *calumniatus* including participial forms like *calumnians* and *calumniatus*; the feminine noun *calumnia* corresponding to the idea of calumny in the abstract; and the masculine noun *calumniator*.

PIE does report three lemmas from this family among the 725 unique to the second-recension *dicta*: *calumnia*, *calumniator*, and *calumniatus*. It does not report the lemma *calumnior* because, as we shall see, the word *calumniantibus* appears in a first-recension *dictum* and, therefore, the verb form is not included in the list of lemmas unique to the second-recension *dicta*.¹⁴

The results reported by PIE are summarized in Table 1 below. The columns indicate:

- the form of the word as it appears in the Friedberg edition,
- the lemmatized form of the word returned by PIE,
- the standard citation for the unit in which the word appears (e.g., C.2, q.3, d.p.c.8),

¹⁴ PIE reports the lemmas as *calumpia*, *calumniator*, and *calumpniatus*. *calumpia* is almost certainly a typo in the LASLA Latin language model for *calumnia*. PIE reports the lemma of *calumpniantibus* as *calumpnio*. The spelling is consistent with the orthographic conventions of the Friedberg edition, from which the text samples of the first- and second-recension *dicta* are ultimately derived.

- the Friedberg edition column number in which the word appears,
- whether the word appears in the first or second recension (e.g., R1 or R2),
- the context (sentence) in which the word appears,
- and the source of the text in which the word appears.

Table 1

word	lemma	location			context	source
<i>calumpniantibus</i>	<i>calumpnio</i>	C.24 q.3 d.p.c.9	993	R1	Ait enim Christus in euangelio: "Orate pro persequentibus et calumpniantibus uos, benefacite his, qui oderunt uos."	Matt 5.44 (quoted from memory; Vulgate order of passage reversed)
<i>calumpniantur</i>	<i>calumpnio</i>	C.23 q.4 d.p.c.23 §3	908	R2	"Quare autem hoc negatum eis fuerit, dicant, si possunt, qui calumpniantur, et ostendant, cur apud eos Dominus mirabilia, quibus profutura non erant, fecerit, et apud eos, quibus erant profutura, non fecerit."	Augustine, <i>De dono perseverantiae</i> (reference to Matt 11.21-24 and/or Luke 10.12-15)
<i>calumpniari</i>	<i>calumpnio</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	"Aut enim calumpniantur, aut preuaricantur, aut tergiuersantur."	Dig. 48.16.1
<i>Calumpniari</i>	<i>calumpnio</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	"Calumpniari est falsa crimina intendere; preuaricari uera crimina abscondere; tergiuersari in uniuersum ab accusatione desistere."	
<i>calumpniari</i>	<i>calumpnio</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	"Sed non utique qui non probat quod intendit protinus calumpniari uidetur."	
<i>calumpnia</i>	<i>calumpnia</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	"Nam eius rei inquisitio arbitrio cognoscentis committitur, qui, reo absoluto, de accusatoris incipit consilio querere, qua mente ductus ad accusationem processit, et si quidem iustum eius errorem reperit, absoluit eum: si uero in euidenti calumpnia eumprehenderit, legitimam penam ei inrogat."	
<i>Calumpniatus</i>	<i>calumpniatus</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	"Nam si quidem ita pronunciauerit: 'Non probasti,' pepercit ei: sin autem pronunciauerit: 'Calumpniatus es,' condemnauit eum."	
<i>calumpniatorem</i>	<i>calumpniator</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	Si autem interloquendo dixerit: Titius temere accusasse uidetur, non calumpniatorem pronunciauit.	Gratian 2 (first-person gloss by author of R2 <i>dicta</i> ; paraphrase of Dig. 48.16.1.5)
<i>calumpniae</i>	<i>calumpnia</i>	C.2 q.3 d.p.c.8 §1	453-454	R2	Temeritas enim facilitatis ueniam continet, et inconsultus calor calumpniae uicio caret, et ob hoc nullam penam huic irrogari oportet.	

When we turn our attention to the substantive treatment of the topic of calumny in the *dicta*, there is variation in terms of the legal sophistication with which the concept is handled, moving generally in the direction of greater technical precision and sophistication. (I say “generally” because, while we can assume the first recension *dicta* were written before the second recension

dicta, we do not have enough information to speculate about the temporal relationship among the second recension *dicta*.)

The concept of calumny makes its initial appearance in the form of a slightly misquoted scriptural reference in the first-recension *dictum* C.24 q.3 d.p.c.9:

Ait enim Christus in euangelio: “Orate pro persequentibus et calumpniantibus uos, benefacite his, qui oderunt uos.”

The treatment of the concept in the second-recension *dictum* C.23 q.4 d.p.c.23 §3 is in a similar spirit, although there the scriptural allusions are mediated through a patristic source, Augustine’s *De dono perseverantiae*, a treatise on predestination.

“Quare autem hoc negatum eis fuerit, dicant, si possunt, qui calumpniantur, et ostendant, cur apud eos Dominus mirabilia, quibus profutura non erant, fecerit, et apud eos, quibus erant profutura, non fecerit.”

Both of these *dicta* use the words associated with the concept of calumny in the same general, non-technical, sense they had in the first millennium of Christian discourse.

That is not the case in the second recension *dictum* C.2 q.3 d.p.c.8 §1. Here we see a series of quotations and a paraphrase from *Dig.* 48.16.1 containing seven occurrences of five words—*calumpniari*, *calummpnia*, *calumpniatus*, *calumpniatorem*, and *calumpniae*—corresponding to

all four of the expected lemmas related to the concept of calumny. It is of obvious interest that the quotations in this *dictum* are directly from Justinianic Roman law.¹⁵

Most interesting of all is that this section of the *dictum* concludes with a first-person saying by the author of the second-recension *dicta* (or at least of this *dictum*) in effect glossing the term *calumniator*.

¹⁵ Notandum quoque est, quod, sicut in Digestis lege XLVIII. legitur, titulo de abolitione criminum: “Accusatorum temeritas tribus modis detegitur. Aut enim calumpniantur, aut preuaricantur, aut tergiuersantur. (*Dig.* 48.16.1.pr.) Calumpniari est falsa crimina intendere; preuaricari uera crimina abscondere; tergiuersari in uniuersum ab accusatione desistere. (*Dig.* 48.16.1.1) Sed non utique qui non probat quod intendit protinus calumpniari uidetur. Nam eius rei inquisitio arbitrio cognoscentis committitur, qui, reo absoluto, de accusatoris incipit consilio querere, qua mente ductus ad accusationem processit, et si quidem iustum eius errorem reperit, absoluit eum: si uero in euidenti calumpnia eum deprehenderit, legitimam penam ei inrogat. (*Dig.* 48.16.1.3) Quorum alterutrum ipsis uerbis pronunciationis manifestatur. Nam si quidem ita pronunciauerit: ‘Non probasti,’ pepercit ei: sin autem pronunciauerit: ‘Calumpniatus es,’ condempnauit eum. Et quamuis nichil de pena subiecerit, tamen legis potestas aduersus eum exercebitur. Nam facti quidem questio in arbitrio est iudicantis: penae uero persecutio non eius uoluntati mandatur, sed legis auctoritati reseruatur. (*Dig.* 48.16.1.4)” Si autem interloquendo dixerit: Titius temere accusasse uidetur, non calumpniatorem pronunciauit. Temeritas enim facilitatis ueniam continet, et inconsultus calor calumpniae uicio caret, et ob hoc nullam penam huic irrogari oportet. (paraphrase of *Dig.* 48.16.1.5)

Si autem interloquendo dixerit: Titius temere accusasse uidetur, non calumpniatorem pronunciauit. Temeritas enim facilitatis ueniam continet, et inconsultus calor calumpniae uicio caret, et ob hoc nullam penam huic irrogari oportet.

Gratian's own intervention is extremely modest—he does little more than paraphrase an opinion attributed to Papinian in *Dig.* 48.16.1.5. Nevertheless, the *dictum* shows progress toward greater legal sophistication in the sense that the discussion draws on resources from Justinianic Roman Law.

Conclusions

The long-term goal of this project has been to find a way to use computationally-enabled distant reading—“reading machines” in the words of Stephen Ramsay¹⁶—to efficiently direct the attention of scholars to specific sites in the text of Gratian's *Decretum* where new topics added between the first and second recensions are likely to be found by close reading.

I think this effort was successful as a proof of concept. Using the PIE lemmatizer in conjunction with the LASLA Latin large language model (LLM) to systematically lemmatize every word in the first- and second-recension *dicta* and then to list all of the lemmas that *do* appear in the second-recension *dicta* but *not* in the first-recension *dicta* drew attention to a family of lemmas (*calumnia*, *calumniator*, and *calumniatus*) that indicated the treatment of a canonically significant concept, calumny. On close reading, the sites in the text of the *dicta* identified by the

¹⁶ Stephen Ramsay, *Reading Machines: Toward an Algorithmic Criticism*, Topics in the Digital Humanities (Urbana: University of Illinois Press, 2011).

results do show a meaningful development over time in the vocabulary of Gratian's *dicta* and to that extent in the teaching or doctrine of the *Decretum* on this topic. The detection of calumny as a topic that the authors of the *Decretum* developed in a significant way in the second recension demonstrates the usefulness of lemmatization as a technique for investigating this type of question.

Calumny was the most obvious topic (at least to me), and I was surprised that there were no other such immediately obvious conceptually related families of lemmas in the results, although I strongly encourage interested readers to examine the complete list of lemmas unique to the second-recension *dicta* for themselves.¹⁷ As previously indicated, there is limited value in the results of machine reading by itself. The real value of the results of machine reading lies in the patterns that researchers see in them.

Is it enough? No.

I have tried to emphasize that the project I have discussed in this paper is very much a work in progress and that the results, although interesting, are limited to the *dicta* and therefore should not be taken as anything more than a proof of concept.

An unsystematic search through the Monumenta Germaniae Historica (MGH) e-text of the Friedberg edition that was created for the *Wortkonkordanz zum Decretum Gratiani* indicates that

¹⁷ The complete list of 725 lemmas unique to the second recension *dicta* is available from my GitHub repository for the Seventeenth International Congress of Medieval Canon Law at <https://github.com/decretist/ICMCL17/blob/main/results/lemmas.txt>.

there are occurrences of forms of the words I have been focusing on—*calumnia*, *calumnior*, and *calumniator*—in the rubrics and canons.¹⁸ A thorough approach to systematically identifying new topics added to the *Decretum* between the first and second recensions will therefore require a data set that includes the rubrics and canons with their inscriptions as well as the *dicta* and case statements.

Ideally, such a data set would be in the form of a new e-text in TEI-P5 XML format incorporating texts from both the old Friedberg edition and the new Winroth edition-in-progress of the first recension. This is where the scale of the undertaking becomes really challenging. Even without the overhead of structuring the data set as a TEI-P5 document, I spent approximately twelve person-weeks on corpus preparation for the *dicta* and the case statements as part of my dissertation project. Since the word count of the canons is roughly five times that of the *dicta*, one person-year is not an unreasonable initial estimate for corpus preparation for a comparable data set for the canons.

The work I have discussed in this paper is based on a highly customized version of a twentieth-century e-text of a nineteenth-century print edition of the *Decretum*. The MGH e-text of the Friedberg edition is the indispensable free resource without which none of my work, and I suspect the work of many others, would be possible. But like so many free things, someone paid

¹⁸ Timothy Reuter and Gabriel Silagi, eds., *Wortkonkordanz zum Decretum Gratiani*, Monumenta Germaniae historica. Hilfsmittel 10 (München: Monumenta Germaniae Historica, 1990). For example, see the rubrics for D.9 c.9 (R1), D.87 c.9 (R2), C.3 q.1 c.6 (R1), and C.5 q.5 c.8 (R2).

a great deal of money to make it free. However, the MGH e-text is a resource that because of its archaic format is approaching the end of its useful life. If we want to continue to advance our understanding of Gratian's *Decretum* with the help of electronic resources, we need to invest time, effort, and grant funding into a twenty-first-century electronic text, or better still an electronic edition, of Gratian's *Decretum*.

Bibliography

Brundage, James A. "The Calumny Oath and Ethical Ideals of Canonical Advocates." In *The Profession and Practice of Medieval Canon Law*, 793–805. Variorum Collected Studies Series CS797. Aldershot: Ashgate Variorum, 2004.

Clérice, Thibault. "Pie Extended, an Extension for Pie with Pre-Processing and Post-Processing." Zenodo, June 2020. <https://doi.org/10.5281/zenodo.3883589>.

Evans, Paul. "Distant Reading of Gratian's *Decretum*." PhD thesis, The Catholic University of America, 2022. <https://jstor.org/stable/community.38760479>.

Manjavacas, Enrique, Ákos Kádár, and Mike Kestemont. "Improving Lemmatization of Non-Standard Languages with Joint Learning." In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 1493–503. Minneapolis, Minnesota: Association for Computational Linguistics, 2019.

McCallum, Andrew Kachites. "MALLET: A Machine Learning for Language Toolkit," 2002.

Pennington, Kenneth. "Gratian and the Jews." *Bulletin of Medieval Canon Law* 31 (2014): 111–24.

———. "The Biography of Gratian, the Father of Canon Law." In *A Service Beyond All Recompense*, edited by Kurt Martens, 359–92. Catholic University of America Press, 2018.

———. "The Law's Violence Against Medieval and Early Modern Jews." *Rivista Internazionale Di Diritto Comune* 23 (2013): 23–44.

Ramsay, Stephen. *Reading Machines: Toward an Algorithmic Criticism*. Topics in the Digital Humanities. Urbana: University of Illinois Press, 2011.

Reuter, Timothy, and Gabriel Silagi, eds. *Wortkonkordanz zum Decretum Gratiani*. Monumenta Germaniae historica. Hilfsmittel 10. München: Monumenta Germaniae Historica, 1990.

Verkerk, Philippe, Yves Ouyard, Margherita Fantoli, and Dominique Longrée. "L.A.S.L.A. And Collatinus: A Convergence in Lexica." *Studi e Saggi Linguistici* 58 (September 2020): 95–120.

Winroth, Anders. *The Making of Gratian's Decretum*. Cambridge: Cambridge University Press, 2000.